

Deploy edge-native camera-LiDAR fusion for real-time roadside perception of vulnerable road users at intersections.

Transportation and Logistics × Worker safety monitoring × Edge computing · OS127 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
24 is the world moving	58 can we play, can we win	MEDIUM 0 named in the evidence	€824m partial evidence · per year	LATER research and standards activi...	L0 1 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity deploys edge-native camera-LiDAR fusion for real-time roadside perception of vulnerable road users at intersections, addressing a deployment gap where existing systems rely on cloud or server-grade infrastructure. It targets transportation and logistics operators who need reliable VRU detection under occlusions, variable lighting, and weather, within strict edge-computing and latency constraints. The opportunity exists now because research has demonstrated a viable edge-deployable framework, but commercial deployment is still nascent.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€1.3bn €290m – €7.7bn	€824m €180m – €4.8bn	€28.9m €6.3m – €168m	partial
Observed: tendered contract value, annualised	€115m	€115m	€4.0m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Transportation and Logistics (EU27_2020)	113,199 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Edge computing	11.3 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC_PCPU	2025
Annual value of one engagement	€792k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Size-weighted buyer base	14,748 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC_PCPU is used as a proxy for Edge computing: Cloud compute purchase as nearest series. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

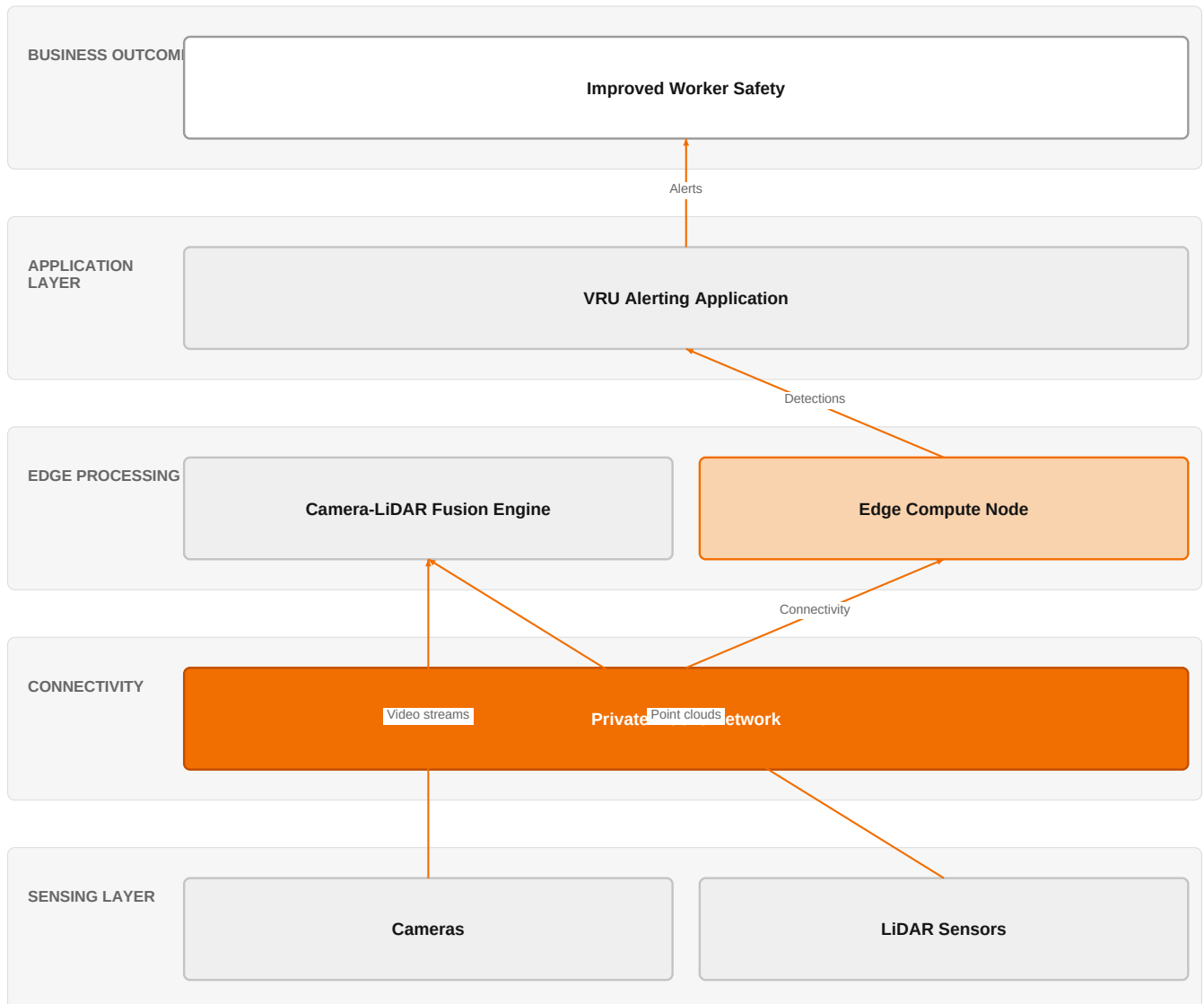
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€115m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in

this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Edge VRU Perception Solution



Orange asset Partner Customer-owned Third party / to be sourced

This solution deploys edge-native camera-LiDAR fusion at intersections to deliver real-time VRU alerts, ensuring safety even under challenging conditions.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Reliable roadside perception of vulnerable road users remains challenging under occlusions, variable lighting, and diverse weather conditions, particularly under strict edge-computing and latency constraints. Existing multi-sensor fusion systems rely on cloud or server-grade infrastructure, creating a deployment gap at real-world intersections. This gap is now being addressed by edge-native frameworks like CLIFE, which enable camera-LiDAR fusion to run on edge devices, making real-time roadside perception feasible at the intersection itself.

Sources: SIG-D768C16F06A7 arxiv.org 2026-07-17

Who buys, and why now

The buyer is likely a safety or operations director at a transportation or logistics company, such as a port operator or a logistics hub manager, who is responsible for worker safety and operational efficiency. The pain is the inability to reliably detect vulnerable road users (e.g., pedestrians, cyclists) near heavy vehicle movements, especially in challenging conditions, leading to safety incidents and operational delays. The trigger is a near-miss or incident report, or a regulatory

push for improved safety at intersections, prompting investment in advanced perception systems.

Why it is hot now — every claim bound to a dated source

- Reliable roadside perception of vulnerable road users remains challenging under occlusions, variable lighting, and weather, especially under strict edge-computing and latency constraints. [SIG-D768C16F06A7]
- Existing multi-sensor fusion systems rely on cloud or server-grade infrastructure, creating a deployment gap at real-world intersections. [SIG-D768C16F06A7]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution combining its Private mobile networks (L0) for reliable, low-latency connectivity at the site, with edge computing infrastructure from partners like NVIDIA, Dell, or HPE to host the camera-LiDAR fusion framework. Orange's IoT experts and Digital experts would integrate sensors and manage the deployment, while Orange Group researchers could provide expertise on the fusion framework. The engagement would start with a site assessment, then deploy edge nodes and sensors, integrate with existing operations, and provide ongoing support.

Why Orange can win

Orange can win by leveraging its Private mobile networks (L0) to ensure reliable, low-latency connectivity that edge computing requires, combined with its IoT and Digital experts for integration and support. The partnership with NVIDIA, Dell, and HPE provides access to proven edge hardware. However, the assets are thin: Orange does not have a proprietary camera-LiDAR fusion framework, and the core technology is still research-stage. Orange's advantage lies in its ability to deliver a complete, managed solution with connectivity and integration, rather than just the perception software.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Private mobile networks	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Dell	partner	L2	partner provides the required technology	unconfirmed
HPE	partner	L2	partner provides the required technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Point One Navigation	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include hyperscalers like AWS, which sells edge computing and competes in Smart Industries; telecom peers like Deutsche Telekom / T-Systems, Lumen Technologies, and Vodafone Business, which sell edge computing and compete in Connectivity Solutions and Smart Industries; global systems integrators like NTT DATA; industrial/OT specialists like Schneider Electric; network equipment vendors like HPE (also a partner) and Nokia; and others. Each brings strengths: AWS has deep cloud and edge services, telecom peers have connectivity and network expertise, and Nokia is active in Transportation and Logistics. The customer will compare on cost, reliability, and integration ease.

Sources: SIG-D768C16F06A7 arxiv.org 2026-07-17

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Edge computing; competes in OX: Smart Industries — also an Orange partner	structural	—

Deutsche Telekom / T-Systems	Telecom operator peer	sells Edge computing; competes in Connectivity Solutions, OX: Smart Industries	structural	—
Lumen Technologies	Telecom operator peer	sells Edge computing; competes in Connectivity Solutions	structural	—
NTT DATA	Global systems integrator	sells Edge computing; competes in Connectivity Solutions, OX: Smart Industries	structural	—
Schneider Electric	Industrial / OT specialist	sells Edge computing; competes in OX: Smart Industries	structural	—
HPE (Aruba, Athonet)	Network equipment vendor	sells Edge computing; competes in Connectivity Solutions — also an Orange partner	structural	—
Nokia	Network equipment vendor	sells Edge computing; active in Transportation and Logistics; competes in Connectivity Solutions, OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	active in Transportation and Logistics; competes in Connectivity Solutions, OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What specific intersections or zones in your operations have the highest risk of VRU incidents, and what is the current mitigation?
- 2 Do you have existing edge computing infrastructure at those sites, or would this require new hardware deployment?
- 3 What are your latency and reliability requirements for real-time perception alerts?
- 4 Have you evaluated any camera-LiDAR fusion systems before, and if so, what were the limitations?
- 5 Who is the primary stakeholder for worker safety improvements, and what is their budget authority?
- 6 Are you currently using a private mobile network or other dedicated connectivity at your sites?

Objections you will meet

They will say	You can answer
This technology is still research-stage; we can't rely on it for safety-critical operations.	You're right that the CLIFE framework is from a research paper, but the underlying components—cameras, LiDAR, edge computing—are mature. We would propose a pilot to validate performance in your environment before any full deployment.
We already have a cloud-based perception system; why move to edge?	Cloud-based systems face latency and reliability issues at the edge, especially in variable conditions. Edge deployment reduces latency and ensures operation even if connectivity drops, which is critical for safety.
This sounds expensive; we have other safety priorities.	The cost is justified by the potential to prevent accidents and improve operational efficiency. We can start with a small pilot to demonstrate value and then scale based on results.

Risks and unknowns

The main risk is that the technology is still at research stage (CLIFE is a framework from an arXiv paper), so commercial readiness is unproven. There is also uncertainty about the regulatory landscape for roadside perception and data privacy, and about whether customers will prioritize this investment over other safety measures. Additionally, the integration with existing infrastructure and the total cost of ownership are not yet clear.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Develop a bid brief for a smart intersection project, outlining a solution that integrates Orange's private mobile networks with NVIDIA, Dell, and HPE edge infrastructure, and leverages Orange Group researchers' expertise to ensure reliable camera-LiDAR fusion under real-world conditions.
Sales	In a conversation with a European city's transportation department, discuss how Orange's private mobile networks combined with edge computing can enable real-time roadside perception for vulnerable road users, referencing the Point One Navigation collaboration as proof of our ability to deliver precise positioning solutions.

Strategist / Innovator	Commission a deep-dive brief on edge-native camera-LiDAR fusion for roadside perception, focusing on how Orange's IoT and Digital experts can leverage NVIDIA, Dell, and HPE edge platforms to address the deployment gap at intersections, and assess the feasibility of a pilot with a transportation authority.
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Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-D768C16F06A7	2026-07-17	arxiv.org	3	CLIFE: Camera-LiDAR Fusion Framework for Edge-Deployable Roadside VRU Perception

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:12+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.